

Pareto-Optimal Urban Transport Systems with Environmental & Health Externalities



Students: Tomas Piskorowski, Abhinav Bhyrraju, Aditya Surendran

School: Athlone Community College

Mentor: Aishling Gillen

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1. Abstract

Urban transport systems represent one of the most complex and pressing challenges in modern city planning, requiring the simultaneous optimisation of environmental sustainability, operational efficiency, and public health. Despite this, many current policy approaches remain narrowly focused—most notably on the transition to electric vehicles—without addressing the broader structural inefficiencies and trade-offs embedded within transport systems.

This project presents a **computational, systems-level analysis** of urban transport design, investigating whether it is possible to achieve strong performance across multiple objectives simultaneously, or whether inherent trade-offs fundamentally limit optimisation. To explore this, a model was developed to simulate **700 realistic urban transport systems**, each defined by varying proportions of petrol cars, electric vehicles, buses, rail, cycling, and walking. These systems were generated within empirically informed constraints to ensure that all configurations reflect plausible real-world conditions.

Each system was rigorously evaluated across five key performance dimensions: CO₂ emissions (g per passenger-kilometre), congestion intensity, average travel speed, physical activity, and pollution exposure. Together, these metrics capture the environmental, efficiency, and health externalities of urban mobility in a unified analytical framework.

To avoid the limitations of single-objective optimisation, this study employs **Pareto optimisation**, a multi-objective method that identifies solutions which cannot be improved in one dimension without compromising another. This approach enables a more

realistic representation of policy decision-making, where competing priorities must be balanced rather than eliminated. The results are striking: only **27 out of 700 systems (3.86%)** were found to be Pareto-optimal, forming a sharply defined efficiency frontier that represents the boundary of achievable performance.

The findings reveal clear structural patterns. Car-dominated systems consistently underperform across multiple objectives, reinforcing the systemic inefficiencies associated with high levels of private vehicle use. While electric vehicles provide measurable reductions in emissions, they fail to address congestion, spatial constraints, and broader system inefficiencies. In contrast, **balanced, multi-modal systems**, characterised by higher shares of public transport and active travel, demonstrate the strongest overall performance across all five metrics.

Critically, the analysis demonstrates that no single transport configuration is universally optimal. Instead, a limited set of high-performing systems exists, each representing a different trade-off between competing objectives. This highlights the importance of **evidence-based, multi-objective decision-making** in urban planning.

Overall, this project provides a robust analytical framework for understanding urban transport systems as interconnected, trade-off-driven structures. The results strongly suggest that meaningful improvements in sustainability and efficiency cannot be achieved through technological substitution alone, but instead require a fundamental shift toward integrated, multi-modal transport strategies.

2. Introduction

Urban transport systems are a central component of modern cities, shaping not only how people move, but also influencing environmental outcomes, economic productivity, and public health. As urban populations continue to grow, the demand for efficient and sustainable transport systems has become increasingly urgent. However, many cities remain heavily dependent on private car use, resulting in a range of interconnected challenges, including rising greenhouse gas emissions, traffic congestion, air pollution, and declining levels of physical activity.

Transport is now recognised as a major contributor to climate change, with road-based transport accounting for a significant proportion of urban emissions. At the same time, congestion reduces economic efficiency by increasing travel times and limiting productivity, while sedentary, car-dependent lifestyles contribute to negative health outcomes. These issues are not isolated; they interact in complex ways, making transport planning a highly challenging, multi-dimensional problem.

In response, policy efforts have increasingly focused on technological solutions, particularly the adoption of electric vehicles. While this transition offers clear benefits in terms of reducing tailpipe emissions, it does not address many of the structural issues inherent in car-dominated transport systems. Electric vehicles still require road space, contribute to congestion, and do not promote physical activity. As a result, there is growing recognition that **technological substitution alone is insufficient** to create truly sustainable urban transport systems.

This highlights the need for a broader, system-level perspective. Rather than evaluating individual transport modes in isolation, it is necessary to consider how entire transport systems function as integrated networks. Different modes—such as public transport, cycling, and walking—interact with one another, and their combined distribution determines overall system performance. Importantly, improving one aspect of a transport system may come at the expense of another, meaning that trade-offs are an unavoidable feature of transport design.

This project adopts a **multi-objective, computational approach** to analysing urban transport systems. By simulating a wide range of possible transport configurations, it seeks to understand how different combinations of transport modes perform across multiple key dimensions. These include environmental impact, system efficiency, and public health outcomes, all of which are critical to sustainable urban development.

A key concept underpinning this study is that of **trade-offs**. In complex systems such as urban transport, it is rarely possible to optimise all objectives simultaneously. Instead, decision-makers must balance competing priorities, making informed compromises based on evidence. This project therefore moves beyond traditional single-metric analysis and instead explores the full landscape of possible solutions.

By applying a structured, data-driven methodology, this study aims to provide deeper insight into the fundamental limitations and opportunities within urban transport design. In doing so, it contributes to a more nuanced understanding of how cities can transition toward transport systems that are not only more sustainable, but also more efficient and supportive of public health.

Ultimately, this project is not focused on promoting a single solution, but rather on understanding the **system-level dynamics** that determine transport performance. This perspective is essential for developing realistic, effective, and evidence-based strategies for the future of urban mobility.

3. Research Question

Urban transport planning is inherently a **multi-objective problem**, in which cities must simultaneously consider environmental sustainability, operational efficiency, and public health outcomes. These objectives are not independent; rather, they are deeply interconnected and often in conflict. For example, policies that prioritise efficiency through increased road capacity may worsen emissions and pollution exposure, while strategies that maximise environmental benefits through active travel may reduce overall travel speed. As a result, transport systems cannot be effectively evaluated using a single metric, and simplistic “one-solution” approaches are unlikely to produce optimal outcomes.

This raises a fundamental question: **is it possible to design an urban transport system that performs well across all major objectives simultaneously, or are trade-offs an unavoidable structural feature of such systems?** Addressing this question requires moving beyond traditional approaches that focus on isolated interventions, such as the adoption of electric vehicles, and instead examining the performance of entire transport systems as integrated, interdependent structures.

The central research question of this project is therefore:

To what extent can urban transport systems be optimised to simultaneously balance environmental impact, congestion, efficiency, and public health, and what trade-offs emerge when attempting to improve these outcomes?

This question is deliberately framed to reflect the **complexity and realism of policy decision-making**, where multiple objectives must

be considered at once. Rather than seeking a single “best” solution, the aim is to understand the range of possible high-performing systems and the compromises that define them.

In order to address this overarching question, several supporting sub-questions are explored:

- **How do different transport modes influence key system outcomes?**

This involves analysing how increasing or decreasing the share of specific modes—such as private cars, public transport, or active travel—affects emissions, congestion, speed, and health-related metrics.

- **Do certain types of transport systems consistently outperform others?**

This focuses on identifying patterns in high-performing systems, particularly whether balanced, multi-modal systems outperform those dominated by a single mode of transport.

- **What is the role of electric vehicles within a broader transport system?**

Given the strong policy emphasis on electrification, this question examines whether electric vehicles meaningfully improve overall system performance beyond emissions reduction.

- **Is there a clear boundary of optimal performance?**

This involves investigating whether a set of systems exists that represents the best possible trade-offs, beyond which improvements in one objective necessarily lead to deterioration in another.

- **How limited are high-performing transport configurations?**

This explores whether optimal systems are common or rare,

providing insight into how constrained urban transport design is in practice.

By addressing these questions, the project seeks to move beyond surface-level conclusions and instead develop a **structured understanding of the trade-offs that define urban transport systems**. The goal is not simply to identify better systems, but to explain *why* they perform better and what this implies for real-world decision-making.

Ultimately, this research is grounded in the recognition that urban transport planning is not a problem of finding a perfect solution, but rather one of navigating competing priorities within a constrained system. Understanding these constraints is essential for developing realistic, evidence-based strategies that can guide cities toward more sustainable, efficient, and healthy transport systems.

4. Methodology

4.1 Overall Approach

This project adopts a **computational, model-based methodology** to analyse urban transport systems as complex, multi-variable systems. Rather than relying on case studies of individual cities, a simulation-based approach was used to systematically explore a wide range of possible transport configurations under controlled conditions.

The objective of this approach is to allow for **structured comparison across many realistic systems**, enabling patterns, trade-offs, and optimal configurations to emerge from the data. By modelling transport systems in a consistent framework, the study avoids the limitations of anecdotal or single-scenario analysis and instead provides a more generalisable understanding of urban transport performance.

4.2 Model Structure and System Representation

Each simulated transport system was defined by a **distribution of transport modes**, representing the proportion of total trips made by each mode. The following six modes were included:

- Petrol cars
- Electric vehicles
- Buses (treated as road-based public transport)
- Rail
- Cycling
- Walking

These modes were selected to capture the major categories of urban transport while maintaining clarity and interpretability within the model.

For each system:

- All mode shares were expressed as percentages
- The total across all modes was constrained to equal **100%**
- This ensured that every system represented a complete and realistic transport structure

This structure allows the model to simulate not just individual modes, but the **interaction between modes**, which is critical in determining overall system performance.

4.3 Generation of Transport Systems

A total of **700 distinct transport systems** were generated for analysis. These systems were not arbitrary, but were constructed within **realistic bounds informed by empirical data**, including European and Irish transport statistics.

The generation process followed these principles:

- Mode shares were varied systematically across plausible ranges
- Extreme or unrealistic combinations were excluded
- The resulting systems reflect **feasible urban transport scenarios**, rather than purely theoretical constructs

The choice of 700 systems represents a balance between:

- Sufficient scale to reveal meaningful patterns
- Computational manageability

- Clear interpretability of results

This approach ensures that the analysis captures a broad design space while remaining grounded in real-world applicability.

4.4 Performance Metrics

Each transport system was evaluated using five key performance indicators, selected to represent the core dimensions of urban transport:

4.4.1 CO₂ Emissions

Emissions were calculated in **grams per passenger-kilometre (g/pkm)** using weighted emission factors for each transport mode.

- Petrol vehicles have high emission factors
- Electric vehicles include indirect emissions from electricity generation
- Public and active transport modes have significantly lower emissions

This metric captures the **environmental impact** of each system.

4.4.2 Congestion Index

Congestion was modelled as a function of total **road-based transport share**, including petrol cars, electric vehicles, and buses.

- Higher road share results in increased congestion
- A nonlinear relationship was applied to reflect real-world traffic behaviour

This metric represents the **efficiency limitations caused by road usage**.

4.4.3 Average Speed

Average system speed was used as a proxy for overall **transport efficiency**.

- Negatively affected by congestion
- Positively influenced by higher rail share

This reflects how quickly people can move through the system under different configurations.

4.4.4 Physical Activity

Physical activity was estimated based on the combined share of **walking and cycling**.

- Higher values indicate greater health benefits
- This serves as a proxy for the role of transport in promoting active lifestyles

4.4.5 Pollution Exposure

Pollution exposure was modelled based on the extent of **road-based transport activity**.

- Higher road usage increases exposure to pollutants
- This metric captures the **public health risks associated with traffic environments**

4.5 Data Sources and Assumptions

The model is informed by a range of real-world data sources, including:

- European Environment Agency (EEA)
- Environmental Protection Agency (EPA)
- Central Statistics Office (CSO)
- National Transport Authority (NTA)
- World Health Organization (WHO)

These sources were used to define realistic parameter ranges and relationships between variables.

Several key assumptions were made:

- Emission factors are constant for each mode
- Electric vehicle emissions include electricity generation impacts
- Buses are treated as road-based vehicles for congestion
- The system operates under **steady-state conditions** (no dynamic traffic fluctuations)

While these assumptions simplify reality, they allow for a **clear and controlled analytical framework**.

4.6 Multi-Objective Optimisation Approach

To evaluate performance across multiple competing objectives, **Pareto optimisation** was applied.

Rather than combining all metrics into a single score, this approach identifies systems that are **non-dominated**, meaning:

A system is Pareto-optimal if no other system performs better in all metrics simultaneously.

This method is particularly suitable for transport analysis because it:

- Avoids subjective weighting of objectives
- Preserves the structure of trade-offs
- Reflects real-world decision-making, where multiple priorities must be balanced

4.7 Identification of Pareto-Optimal Systems

All 700 systems were compared against each other using Pareto dominance criteria.

- Systems that were strictly worse across all metrics were eliminated
- Only systems that could not be improved without trade-offs were retained

As a result:

- **27 systems (3.86%)** were identified as Pareto-optimal

These systems form the **Pareto frontier**, representing the boundary of optimal performance within the model.

4.8 Analytical Strength of the Method

This methodology provides several key strengths:

- It captures **system-wide interactions**, not just individual variables
- It enables **objective comparison across many scenarios**
- It reveals **hidden trade-offs and structural constraints**
- It produces results that are both **quantitative and interpretable**

Overall, the approach provides a **robust, evidence-based framework** for analysing urban transport systems, allowing for deeper insight into how different transport strategies perform under realistic conditions.

5. Results

5.1 Overview of Simulation Outcomes

A total of **700 urban transport systems** were generated and evaluated using the five performance metrics defined in the methodology: CO₂ emissions, congestion, average speed, physical activity, and pollution exposure. Each system represents a distinct combination of transport modes, allowing for a comprehensive exploration of the possible design space.

The analysis revealed that the vast majority of systems perform sub-optimally when evaluated across multiple objectives simultaneously. Only **27 systems (3.86%)** were identified as Pareto-optimal, meaning that they cannot be improved in one metric without worsening at least one other.

This result highlights the **highly constrained nature of effective transport system design**, where only a small fraction of realistic configurations achieve strong overall performance.

5.2 Trade-Offs Between Key Objectives

The correlation analysis (Graph 1) provides insight into the relationships between key performance metrics across all 700 systems.

Several strong patterns are observed:

- **Emissions and congestion show a positive correlation**, indicating that systems with higher reliance on road-based

transport tend to perform poorly in both environmental and efficiency terms

- **Congestion and average speed exhibit a strong negative relationship**, reflecting the expected impact of traffic density on system efficiency
- **Physical activity is often negatively correlated with speed**, as systems that prioritise walking and cycling typically involve slower modes of transport

These relationships demonstrate that improvements in one objective frequently come at the expense of another, confirming that **trade-offs are a fundamental characteristic of urban transport systems** rather than an exception.

5.3 Structure of the Pareto Frontier

The Pareto frontier (Graph 4) represents one of the most significant outputs of the analysis. This graph plots all 700 systems in terms of emissions and congestion, with Pareto-optimal systems forming a distinct boundary.

Key observations include:

- The Pareto-optimal systems define a **clear lower boundary**, representing the best achievable combinations of low emissions and low congestion
- Systems located above this boundary are **dominated**, meaning there exist alternative configurations that perform better across both metrics

- Movement along the frontier reflects **trade-offs between emissions and congestion**, where reducing one often leads to increases in the other

This boundary represents the **limit of achievable performance within the model**, illustrating that beyond a certain point, improvements in one objective are only possible through compromise in another.

5.4 Diversity of Pareto-Optimal Systems

Further analysis of the 27 Pareto-optimal systems reveals that they are not uniform, but instead form **distinct clusters of performance strategies** (Graph 2).

These clusters can be broadly categorised as:

Efficiency-Oriented Systems

- Characterised by higher rail usage
- Achieve higher average speeds
- Maintain moderate emissions and congestion levels

Health and Low-Emission Systems

- High levels of walking and cycling
- Low emissions and pollution exposure
- Reduced average speeds due to reliance on active travel

Balanced Systems

- Moderate performance across all five metrics
- Avoid extreme trade-offs
- Represent the most stable and adaptable configurations

This diversity demonstrates that there is **no single optimal transport system**, but rather a set of high-performing systems, each representing a different balance of priorities.

5.5 Trade-Off Behaviour Within Optimal Systems

The internal structure of the Pareto-optimal systems is further illustrated using the parallel coordinate plot (Graph 3), where each line represents one of the 27 systems across all five metrics.

This analysis shows that:

- No system achieves maximum performance in all metrics simultaneously
- Systems that minimise emissions may accept lower speeds
- Systems that maximise speed may tolerate higher congestion or emissions
- Balanced systems maintain intermediate values across all dimensions

This reinforces the conclusion that **all optimal systems involve compromise**, and that performance improvements are inherently limited by competing objectives.

5.6 Sensitivity Analysis: Drivers of System Performance

The sensitivity analysis (Graph 5) identifies how changes in transport mode share influence system outcomes.

Key findings include:

- **Petrol car usage strongly increases emissions**, confirming its dominant environmental impact
- **Electric vehicles reduce emissions**, but continue to contribute to congestion due to their reliance on road infrastructure
- **Rail has a strong positive effect on speed**, improving overall system efficiency
- **Walking and cycling significantly increase physical activity**, contributing to improved health outcomes
- **All road-based modes contribute to higher congestion and pollution exposure**, regardless of fuel type

This analysis highlights that **mode choice is a critical driver of system performance**, and that structural changes in transport composition are necessary to achieve meaningful improvements.

5.7 Interpretation of the Pareto-Optimal Table

The full set of 27 Pareto-optimal systems is presented in tabular form (Appendix E), where each row represents a distinct transport configuration.

The table shows:

- The percentage distribution of transport modes for each system
- The corresponding performance across all five metrics
- Classification into clusters based on performance characteristics

One system is highlighted as the **most balanced configuration**, demonstrating relatively strong performance across all objectives without extreme trade-offs.

This table provides concrete evidence that multiple viable transport strategies exist, each representing a different approach to balancing competing priorities.

5.8 Key Results Summary

The analysis produces several clear and consistent conclusions:

- Only a small proportion of systems perform well across all objectives
- Car-dominated systems consistently underperform
- Electric vehicles improve emissions but do not resolve congestion or system inefficiencies
- Multi-modal systems with strong public transport and active travel components achieve the best overall performance
- Trade-offs between objectives are unavoidable and define system behaviour

5.9 Overall Interpretation

Taken together, the results demonstrate that urban transport systems are governed by **structural constraints and trade-offs**. The presence of a clearly defined Pareto frontier indicates that there are limits to what can be achieved within a given system configuration.

Rather than identifying a single optimal solution, the analysis reveals a **set of efficient trade-offs**, each corresponding to different policy priorities. This highlights the importance of adopting a multi-objective perspective in transport planning and avoiding overly simplistic solutions.

6. Discussion

6.1 Interpreting the Core Findings

The results of this study demonstrate that urban transport systems cannot be optimised across all key objectives simultaneously without encountering trade-offs. The identification of only **27 Pareto-optimal systems out of 700** highlights how limited high-performing configurations are within a realistic design space. This suggests that effective transport systems are not simply a matter of selecting the “right” solution, but rather of navigating a constrained set of viable trade-offs.

A key insight is that **system performance is driven more by overall structure than by individual technologies**. Transport systems dominated by private cars consistently perform poorly across multiple metrics, regardless of whether those cars are petrol or electric. This indicates that inefficiencies are not solely a function of emissions, but are embedded in the broader characteristics of road-based transport, including congestion and land use.

6.2 Limitations of Single-Solution Approaches

One of the most significant findings is the limited effectiveness of electric vehicles as a standalone solution. While electric vehicles clearly reduce emissions, the results show that they do not address congestion, average speed, or pollution exposure to the same extent.

This reflects a broader issue in transport policy: the tendency to prioritise **technological substitution over structural change**. By focusing primarily on replacing petrol vehicles with electric ones,

policies may overlook deeper inefficiencies associated with car-dependent systems. The model demonstrates that even fully electrified car-heavy systems remain suboptimal when evaluated across multiple objectives.

This finding does not suggest that electric vehicles are unimportant, but rather that their impact is **partial and context-dependent**, and must be considered within the wider transport system.

6.3 Importance of Multi-Modal Transport Systems

The strongest-performing systems identified in the analysis are consistently **multi-modal**, incorporating a balanced mix of public transport and active travel. These systems achieve relatively strong performance across all five metrics, without extreme weaknesses in any single area.

Public transport, particularly rail, plays a key role in improving efficiency by enabling higher travel speeds and reducing reliance on congested road networks. At the same time, walking and cycling contribute significantly to improved health outcomes and reduced emissions.

This combination allows balanced systems to **mitigate trade-offs**, rather than eliminate them entirely. While no system achieves optimal performance in all dimensions, multi-modal systems come closest to achieving a practical and sustainable balance.

6.4 Understanding Trade-Off Structures

The presence of a clearly defined Pareto frontier demonstrates that urban transport systems are governed by **structural trade-offs**. These trade-offs arise because different transport modes inherently prioritise different outcomes:

- Road-based transport offers flexibility but increases congestion and emissions
- Active transport improves health but reduces speed
- High-speed systems often require infrastructure that limits accessibility or increases cost

As a result, improving one objective often requires accepting compromises in another. The parallel coordinate analysis further reinforces this, showing that each Pareto-optimal system represents a different combination of strengths and weaknesses.

This highlights that **there is no universally optimal transport system**, only systems that are optimal under specific priorities.

6.5 Implications for Decision-Making

The findings of this study have important implications for real-world transport planning. Rather than attempting to maximise a single metric, policymakers must adopt a **multi-objective perspective**, recognising that trade-offs are unavoidable.

This requires:

- Clearly defining policy priorities (e.g. emissions vs efficiency vs health)

- Understanding the consequences of prioritising one objective over another
- Evaluating systems as integrated wholes rather than isolated components

The use of Pareto optimisation provides a useful framework for this process, as it allows decision-makers to identify a set of efficient options rather than a single “best” solution.

6.6 Strengths of the Modelling Approach

A key strength of this study is its ability to analyse a large number of transport systems within a consistent framework. By generating 700 realistic configurations, the model provides a broad and systematic exploration of the transport design space.

The use of multiple performance metrics ensures that results are not biased toward a single objective, while Pareto optimisation allows for a transparent and rigorous evaluation of trade-offs.

Importantly, the model prioritises **clarity and interpretability**, making it possible to identify clear patterns and relationships between variables. This makes the findings more accessible and applicable to real-world decision-making.

6.7 Conceptual Contribution of the Study

Beyond its specific results, this project contributes to a broader understanding of urban transport systems as **complex, interdependent systems**. It demonstrates that transport challenges

cannot be solved through isolated interventions, but instead require coordinated, system-level strategies.

The study also highlights the importance of **evidence-based analysis** in transport planning. By using a structured modelling approach, it becomes possible to move beyond assumptions and evaluate policies based on their actual system-wide impacts.

6.8 Overall Interpretation

Overall, the discussion reinforces a central conclusion: urban transport systems are defined by **constraints and trade-offs**, and effective solutions must work within these limits rather than attempt to eliminate them.

The results suggest that the most promising approach to sustainable urban transport lies in **balanced, multi-modal systems**, supported by integrated planning and multi-objective analysis. This represents a shift away from simplistic, single-solution thinking and toward a more realistic and comprehensive understanding of urban mobility.

7. Real-World Applications and Impact

7.1 Translating Results into Real-World Context

While this study is based on a computational model, its findings have clear and direct relevance to real-world urban transport planning. The model was intentionally designed using realistic constraints and data-informed assumptions, meaning the patterns observed are not purely theoretical, but reflect structural behaviours that are likely to occur in actual cities.

The results demonstrate that transport performance is not determined by a single factor, but by the **overall composition of the system**. This insight is critical for policymakers, as it suggests that focusing on isolated interventions—such as promoting electric vehicles—may fail to produce meaningful improvements if the broader system remains unchanged.

7.2 Implications for Urban Transport Policy

The findings highlight several key principles that can inform transport policy:

Reducing Car Dependency

Car-dominated systems consistently underperform across multiple objectives, including emissions, congestion, and health outcomes. This suggests that policies aimed at reducing reliance on private vehicles—such as limiting car use in city centres or improving alternatives—are likely to produce significant benefits.

Prioritising Public Transport

Public transport, particularly rail, plays a crucial role in improving system efficiency. By moving large numbers of people without increasing road congestion, it provides a scalable solution for urban mobility.

Promoting Active Travel

Walking and cycling contribute directly to improved public health while also reducing emissions and pollution exposure. Encouraging active travel through infrastructure and urban design can therefore deliver multiple benefits simultaneously.

Moving Beyond Single-Solution Strategies

The limited impact of electric vehicles on congestion highlights the need for **integrated policy approaches**. Effective transport strategies must consider how different modes interact, rather than treating them as independent solutions.

7.3 Application to Real Cities

The insights from this project are particularly relevant to cities with high levels of car dependency, such as **Dublin**.

In such contexts:

- Increasing public transport capacity could reduce congestion and improve efficiency
- Expanding cycling and walking infrastructure could improve health outcomes

- Reducing car usage in urban centres could significantly lower emissions

While the model does not replicate any specific city exactly, it provides a **framework for evaluating potential transport strategies** in real-world settings.

7.4 Quantitative Impact Potential

Although the model is not designed to produce exact real-world predictions, it allows for indicative estimates of impact:

- Shifting from car-heavy systems to balanced systems could result in **substantial reductions in CO₂ emissions**
- Reductions in road usage could lead to **significant decreases in congestion levels**
- Increased active travel could produce measurable improvements in **population-level physical activity**

These outcomes demonstrate that **system-level changes can produce multiple benefits simultaneously**, rather than requiring trade-offs in all cases.

7.5 Broader Societal Impact

Beyond transport efficiency, the findings have wider implications:

- **Environmental impact:** Lower emissions contribute to climate change mitigation
- **Public health:** Increased physical activity and reduced pollution improve overall health outcomes

- **Urban quality of life:** Reduced congestion and cleaner environments enhance livability

This reinforces the idea that transport systems are not just infrastructure, but **key determinants of urban sustainability and well-being**.

7.6 Alignment with Global Goals

The results of this project align with key objectives outlined by the United Nations Sustainable Development Goals (SDGs), particularly:

- Sustainable Cities and Communities
- Climate Action
- Good Health and Well-being

By demonstrating how transport systems can be optimised across these dimensions, the project contributes to a broader global effort toward sustainable development.

7.7 Practical Use of the Model

The modelling framework developed in this project could be adapted for practical use by:

- Urban planners evaluating transport strategies
- Policymakers comparing different policy scenarios
- Researchers studying multi-objective optimisation in transport systems

With further development, the model could be extended to incorporate real-world data, allowing for **city-specific analysis and decision support**.

7.8 Overall Impact

Overall, this project demonstrates that meaningful improvements in urban transport systems require **system-level thinking and integrated planning**. The results provide clear evidence that balanced, multi-modal systems offer the most effective pathway toward achieving environmental, efficiency, and health objectives simultaneously.

By moving beyond single-solution approaches and embracing multi-objective analysis, cities can make more informed, evidence-based decisions about the future of urban mobility.

10. Conclusion

This project set out to investigate whether urban transport systems can be designed to perform well across multiple critical objectives—environmental sustainability, operational efficiency, and public health—or whether trade-offs between these objectives are unavoidable.

Through the development and analysis of a computational model simulating **700 realistic transport systems**, the study provides a clear and evidence-based answer: **trade-offs are an inherent and unavoidable feature of urban transport design**. Only **27 systems (3.86%)** were identified as Pareto-optimal, demonstrating that high-performing configurations are both limited and structurally constrained.

A central finding of the project is that **system composition matters more than individual technologies**. Transport systems dominated by private cars consistently underperform across multiple metrics, regardless of whether those vehicles are petrol or electric. While electric vehicles contribute to emissions reduction, they do not address congestion, road space limitations, or broader system inefficiencies. This highlights the limitations of single-solution approaches and reinforces the need for more comprehensive strategies.

In contrast, the most effective systems identified in the analysis are **balanced, multi-modal systems**, combining public transport with active travel. These systems achieve strong overall performance by distributing demand across different modes, reducing pressure on road infrastructure, and delivering co-benefits in terms of emissions, efficiency, and public health.

Importantly, the study demonstrates that there is **no single “best” transport system**, but rather a set of optimal solutions, each representing a different balance of competing priorities. This reflects the reality of urban planning, where decisions must be made within constraints and trade-offs must be actively managed rather than avoided.

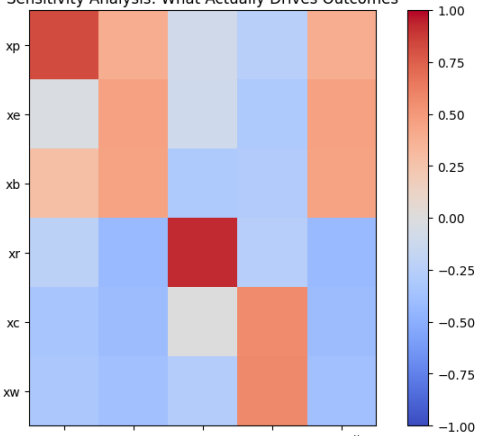
The use of Pareto optimisation provides a powerful framework for understanding these trade-offs, allowing for the identification of efficient solutions without imposing arbitrary assumptions about the relative importance of different objectives. This approach aligns closely with real-world decision-making, where multiple outcomes must be considered simultaneously.

Overall, this project contributes to a deeper understanding of urban transport systems as **complex, interdependent, and trade-off-driven structures**. It demonstrates that meaningful improvements cannot be achieved through technological substitution alone, but require a shift toward **integrated, system-level thinking**.

The findings provide clear evidence that reducing car dependency and promoting multi-modal transport systems offers the most effective pathway toward sustainable, efficient, and healthy urban mobility. As cities continue to face growing environmental and societal pressures, such approaches will be essential in shaping the future of urban transport.

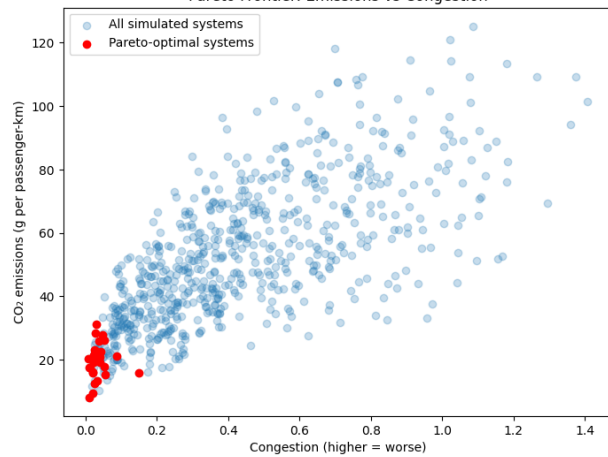
Red = increasing this mode increases the outcome.
Blue = increasing this mode reduces the outcome.

Sensitivity Analysis: What Actually Drives Outcomes



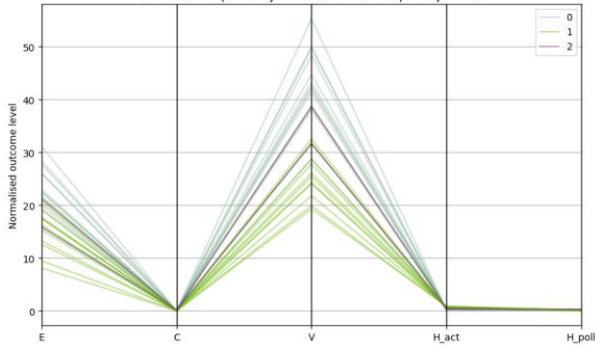
Each point is a possible urban transport system.
Red points cannot reduce emissions without increasing congestion (or vice versa).

Pareto Frontier: Emissions vs Congestion



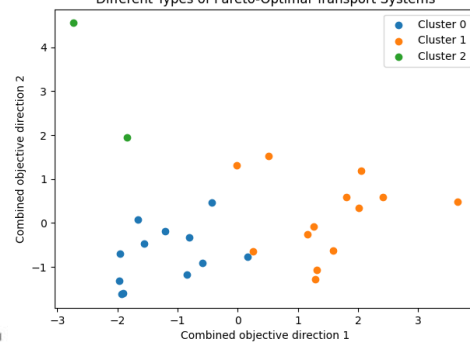
Each line is one Pareto-optimal system.
No line is best on all axes; this is why Pareto analysis is required.

How Pareto-Optimal Systems Trade Off Multiple Objectives



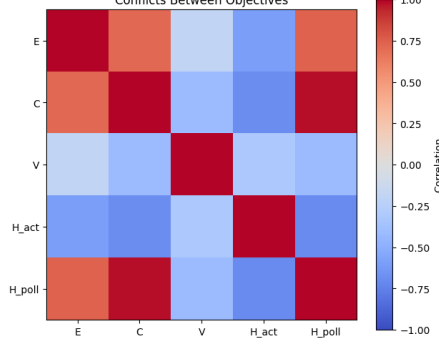
Each cluster represents a distinct policy strategy
(e.g. low-emissions, high-efficiency, or balanced systems).

Different Types of Pareto-Optimal Transport Systems



Strong negative values show trade-offs.
This explains why no single transport solution is 'best'.

Conflicts Between Objectives



No.	Petrol cars (%)	Electric cars (%)	Buses (%)	Rail (%)	Cycling (%)	Walking (%)	CO ₂ (g/pkm)	Congestion	Speed (km/h)	Physical activity	Pollution exposure	Cluster
58	12	4	2	22	51	9	27.8	0.047	28.8	0.60	0.18	0
59	4	0	3	41	37	16	20.4	0.007	28.7	0.53	0.07	1
123	12	1	1	54	10	22	31.1	0.029	48.3	0.32	0.14	0
153	1	8	6	63	15	8	22.5	0.032	55.3	0.23	0.15	0
177	1	4	6	30	59	1	16.0	0.020	31.6	0.59	0.11	1
185	3	5	5	48	24	15	21.6	0.026	42.5	0.39	0.13	0
215	4	6	2	46	33	9	16.3	0.021	38.6	0.42	0.12	1
218	1	9	2	16	36	36	9.5	0.020	24.0	0.73	0.11	1
263	4	3	5	40	36	12	19.8	0.021	38.0	0.48	0.12	0
306	4	10	3	65	0	18	20.9	0.041	50.0	0.18	0.16	0
311	3	5	0	1	85	6	8.1	0.010	19.3	0.91	0.08	1
366	6	0	1	37	25	30	17.5	0.009	32.5	0.56	0.08	1
389	1	13	2	57	20	7	19.1	0.040	49.7	0.27	0.16	0
422	10	2	0	16	49	23	21.5	0.022	26.1	0.72	0.12	1
428	12	1	3	34	31	19	26.0	0.038	38.6	0.50	0.16	0
443	8	2	2	58	26	3	23.0	0.024	41.4	0.29	0.13	0
444	9	2	1	8	64	16	19.2	0.021	19.9	0.80	0.12	1
457	6	7	1	70	1	16	28.4	0.027	44.8	0.17	0.13	0
458	3	12	4	29	42	11	15.3	0.054	31.7	0.52	0.19	1
556	5	12	1	40	6	34	26.1	0.053	42.1	0.41	0.19	0
569	4	9	4	44	7	33	22.5	0.043	43.2	0.40	0.17	0
612	0	6	13	15	47	19	17.7	0.053	25.4	0.66	0.19	1
637	0	28	3	34	16	18	15.8	0.150	31.7	0.35	0.32	2
650	4	7	4	5	77	4	13.2	0.032	21.8	0.81	0.15	1
667	5	4	0	30	42	19	17.5	0.012	27.8	0.61	0.09	1
675	5	19	1	30	45	1	21.2	0.088	38.8	0.46	0.24	2
684	3	9	0	21	2	64	12.6	0.025	24.3	0.66	0.13	1